1. INTRODUCTION

1.1 Project Overview

This project focuses on analyzing housing market trends using Tableau for visualization and Flask for web integration. The housing dataset contains details such as sale price, renovation year, number of bedrooms, bathrooms, floors, basement area, and house age.

The objective of this project is to transform raw housing data into interactive visual insights through dashboards and storyboards. The final dashboard and story are embedded into a web application using Flask.

1.2 Purpose

- To analyze housing sale price trends
- To study renovation impact on sales
- To understand house age distribution
- To create interactive dashboard and story
- To integrate Tableau dashboard with Flask UI

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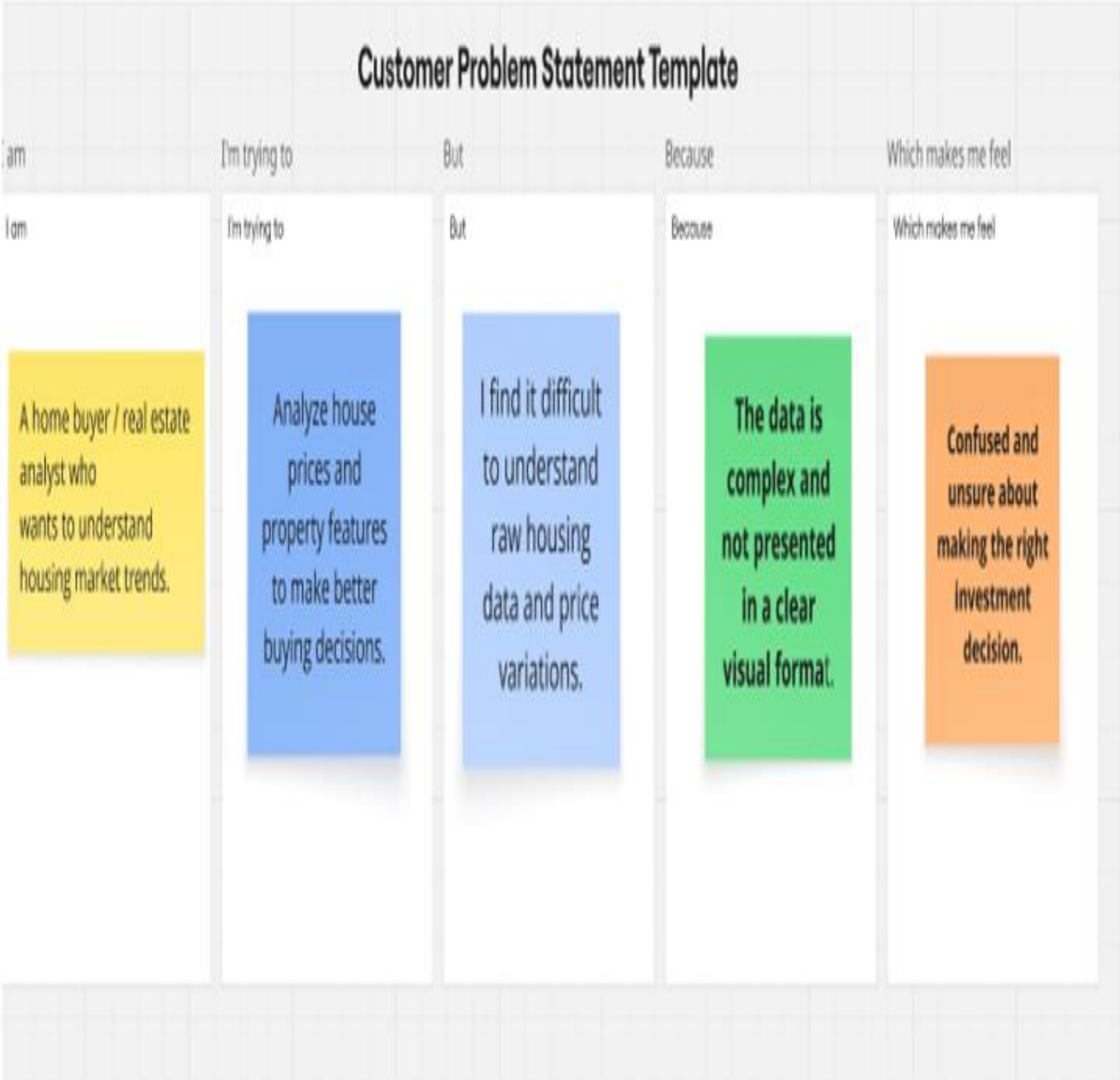
2. IDEATION PHASE

2.1 Problem Statement

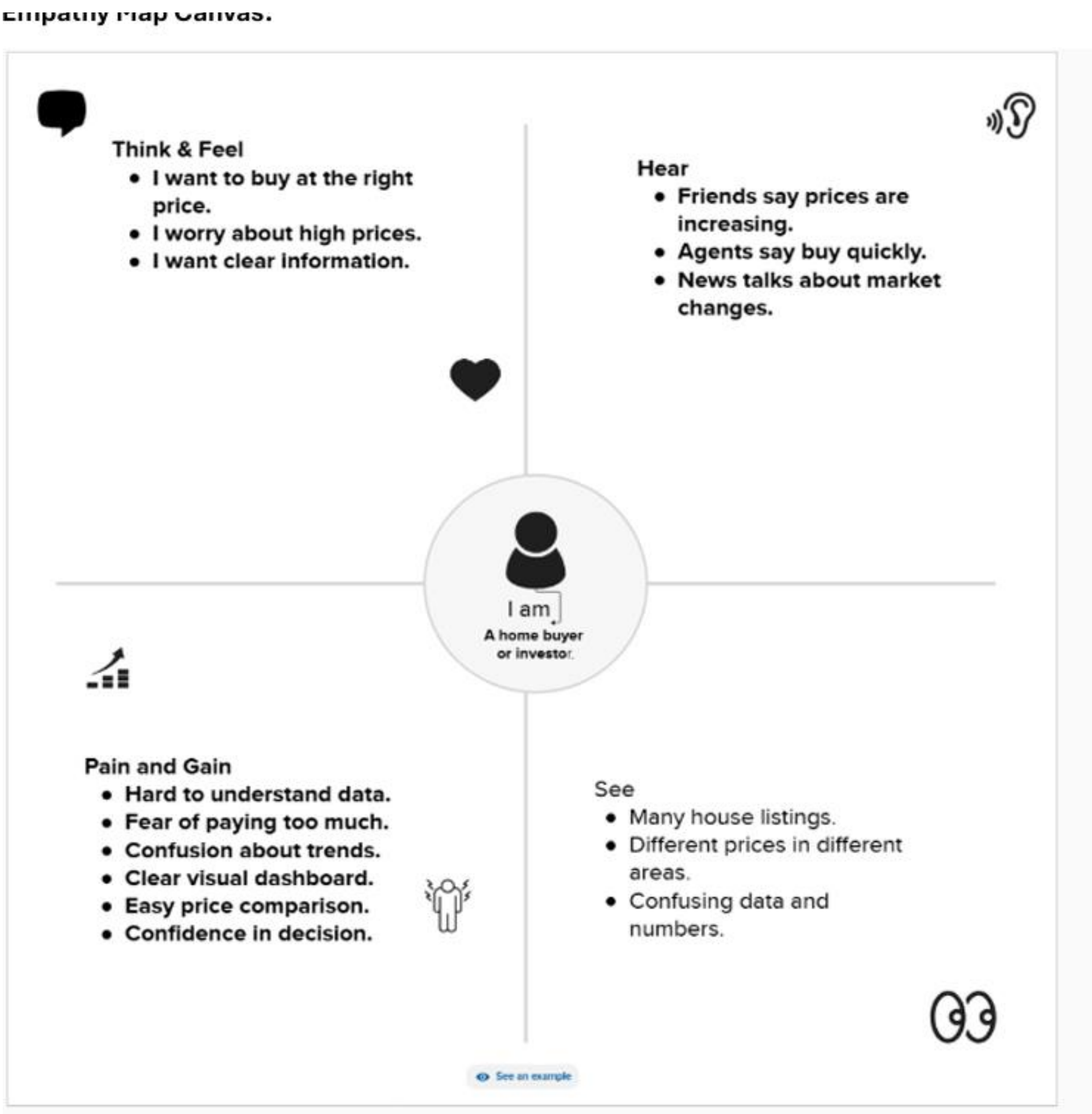
Understanding housing market data is difficult when presented in raw tabular format. Buyers, sellers, and investors require clear visual insights to analyze:

- Sale price trends
- Renovation impact
- House age distribution
- Effect of bedrooms, bathrooms, and floors

Therefore, a visualization-based solution is required.


Customer Problem Statement Template
I am
I'm trying to
But
Because
Which makes me feel
I am
I'm trying to
But
Because
Which makes me feel
A home buyer / real estate analyst who wants to understand housing market trends.
Analyze house prices and property features to make better buying decisions.
I find it difficult to understand raw housing data and price variations.
The data is complex and not presented in a clear visual format.
Confused and unsure about making the right investment decision.

2.2 Empathy Map Canvas



2.3 Brainstorming

Ideas discussed:

- Create sales overview visualization
- Show average sale price
- Show renovation impact
- Analyze house age
- Build interactive dashboard
- Integrate with Flask

Final Idea:
👉 Tableau Dashboard + Story + Flask Integration

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3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. Entice (Awareness)	2. Enter (Access Website)	3. Engage (Explore Dashboard)	4. Exit (Decision Making)	5. Extend (After Usage)
Steps • User wants to understand houing prices	· User wants to understand houing prices · Searches for market trends	· Opens Flask web application · Views embedded Tableau dashboard	· Analyzes trends. · Makes buying/ investment decision	· Shares dashboard · Revisits for future analysis
Goal	· Get clear price insights	· Explore housing data	· Explore housing data	· Track market changes
Pain Pain (Challenges)	· Raw data is confusing	· Too many charts may confuse	· Clean layout and clear headings	· Add chart layout and headings
Opportunity	· Provide simple dashboard homepage	· Clean layout and clear headings	· Add chart descriptions and tooltips	· Update dataset regularly
Opportunity	· Provide simple dashboard homepage	· Clean layout and clear headings	· Add chart descrip-tions and tooltips	· Add download/share feature

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3.2 Solution Requirement

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Form (Email & Password)
		Registration through Gmail
		Registration through LinkedIn
FR-2	User Confirmation	Confirmation via Email
		Confirmation via OTP
		Account activation after verification
FR-3	User Login & Authentication	Login using Email & Password
		Login using Gmail
		Forgot Password option
		Logout functionality
FR-4	Dashboard Access	Access dashboard after login
		View housing price trends
		View renovation impact analysis
		View sales distribution charts
FR-5	Data Filtering	Filter by bedrooms
		Filter by location
		Filter by renovation status
		Filter by price range
FR-6	Data Visualization	Interactive Tableau dashboard embedding
		Real-time chart updates
FR-7	Admin Management	Upload housing dataset
		Manage registered users
		Update/Delete dataset

Non-functional Requirements:

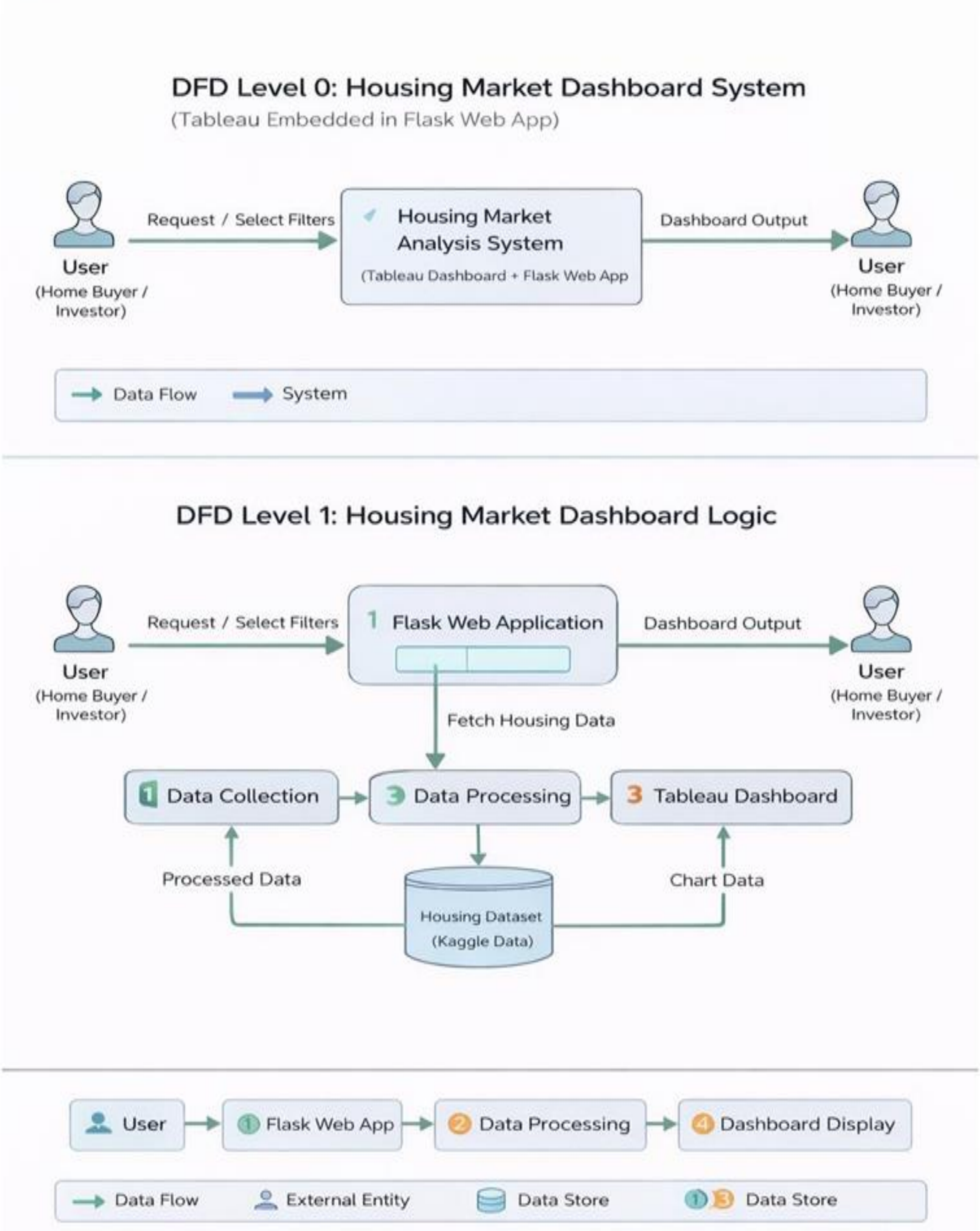
Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	Users can easily view and filter housing dashboard.
NFR-2	**Security**	Login system protects user accounts and data.
NFR-3	**Reliability**	Dashboard shows correct housing price data without errors.
NFR-4	**Performance**	Tableau dashboard loads fast inside Flask app.
NFR-5	**Availability**	Web application is accessible anytime online
NFR-6	**Scalability**	System supports more housing data and users in future.

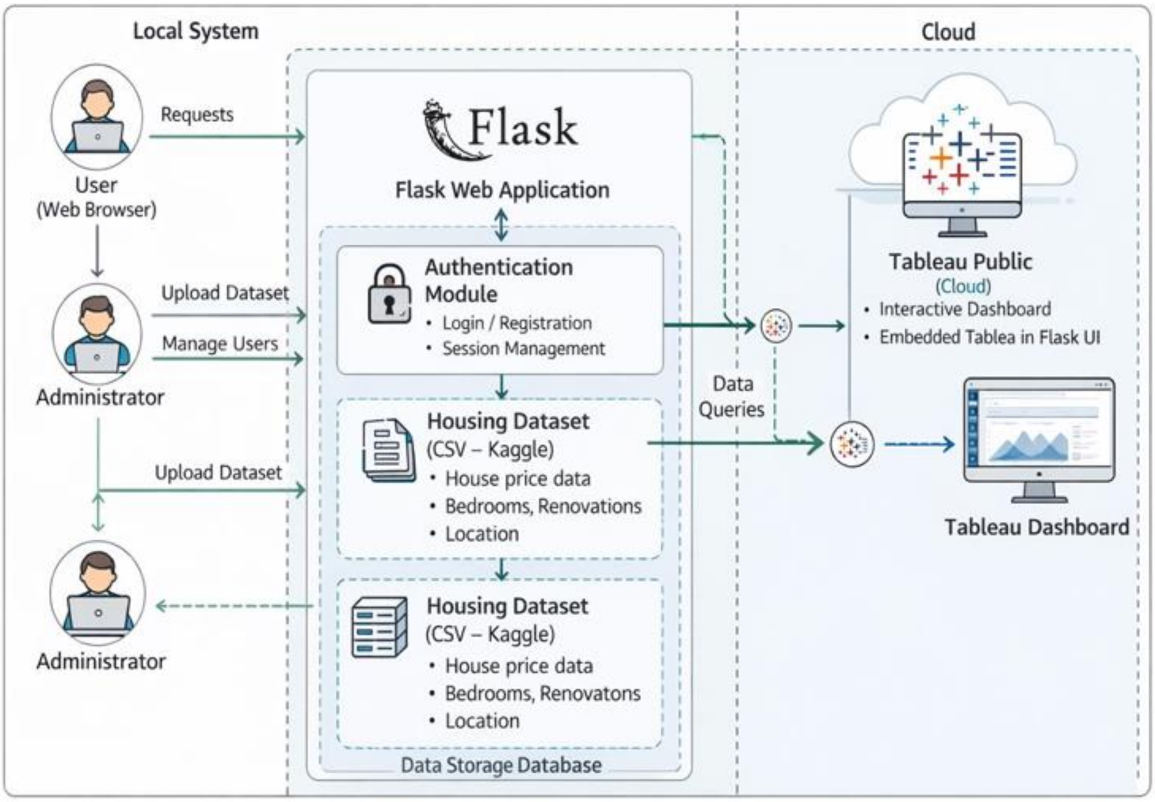
3.3 Data Flow Diagram

Data Flow Diagrams:

Flow



3.4 Technology Stack



S.No	Component	Description	Technology
1	User Interface	Web UI where user logs in and views dashboard	HTML, CSS, Bootstrap
2	Application Logic-1	Handles login, routing and request processing	Python (Flask)
3	Application Logic-2	Data filtering and processing logic	Python (Pandas)
4	Application Logic-3	Embedding and displaying Tableau dashboard	Tableau Public (Embed Code)
5	Database	Stores user login details	SQLite
6	Cloud Database	Not used in this project	Not Applicable
7	File Storage	Stores housing dataset CSV file	Local File System
8	External API-1	Interactive dashboard visualization service	Tableau Public
9	External API-2	Gmail login (if implemented)	Google OAuth API
10	Machine Learning Model	Not used in this system	Not Applicable
11	Infrastructure (Server / Cloud)	Application runs locally and connects to Tableau Cloud	Local Server (Flask) + Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Backend framework used for development	Flask (Python)
2	Security Implementations	Secure login, password hashing, session control	Flask Session, Werkzeug Security
3	Scalable Architecture	Modular design allowing future data expansion	Flask Architecture
4	Availability	Accessible when server is running	Local Server / Cloud Hosting
5	Performance	Fast dashboard loading and efficient filtering	Pandas + Tableau Rendering

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4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem:
Raw housing data is difficult to interpret.

Solution:
Interactive Tableau dashboard and story embedded in Flask.

4.2 Proposed Solution

The system:

- Downloads housing dataset
- Cleans and prepares data
- Creates 6 unique visualizations
- Builds dashboard
- Creates storyboard
- Publishes to Tableau Public
- Embeds dashboard in Flask UI

4.3 Solution Architecture

User
↓
Flask Web Application
↓
Embedded Tableau Dashboard

↓
Tableau Public Server

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5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Activity
Phase 1	Data Collection
Phase 2	Data Preparation
Phase 3	Data Visualization
Phase 4	Dashboard Creation
Phase 5	Story Creation
Phase 6	Performance Testing
Phase 7	Web Integration
Phase 8	Documentation

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6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Amount of Data Loaded

- Housing dataset loaded successfully

Utilization of Filters

- Sale Price – Top N Filter
- Age of House – Top N Filter

Number of Calculation Fields

- House Age (Calculated Field)

Number of Visualizations

1. Count of Transformed Housing Data
2. Total Sales by Years Since Renovation
3. Distribution of House Age by Renovation Status
4. House Age Distribution by Bathrooms, Bedrooms, and Floors

Total Unique Visualizations: **4**

7. RESULTS

7.1 Output Screenshots

(Insert the following screenshots)

- Visualization Sheets

- Dashboard
- Story
- Filters
- Flask Web UI

8. ADVANTAGES & DISADVANTAGES

Advantages

- Easy understanding of housing trends
- Interactive filters
- Web-based accessibility
- Clear data insights

Disadvantages

- Depends on Tableau Public server
- Requires internet connection

9. CONCLUSION

The project successfully visualizes housing market trends using Tableau and integrates the dashboard with Flask. The system helps users analyze sale price patterns, renovation impact, and house age distribution in an interactive way.

10. FUTURE SCOPE

- Add Machine Learning prediction model
- Real-time data integration
- Cloud deployment
- User authentication system

11. APPENDIX

- ### Source Code:

❖ **Flask code:**

```
from flask import Flask, render_template


app = Flask(__name__)


@app.route('/')
def home():
   return render_template('index.html')


if __name__ == '__main__':
   app.run(debug=True)
```

❖ **Index.htnl [html code]:**

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Housing Market Trends</title>
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<style>
html { scroll-behavior: smooth; }

body {
  font-family: 'Segoe UI', sans-serif;
}
```

```css
/* Navbar */
.navbar {
  background: #0d6efd;
}

.navbar .nav-link,
.navbar-brand {
  color: white !important;
  font-weight: 500;
}

/* Sections Animation */
section {
  min-height: 100vh;
  padding: 100px 0;
  opacity: 0;
  transform: translateY(40px);
  transition: 0.8s ease-in-out;
}

section.show {
  opacity: 1;
  transform: translateY(0);
}

/* Home */
.home {
  background: linear-gradient(to right, #4e73df, #1cc88a);
  color: white;
  text-align: center;
}

/* Buttons */
.btn-custom {
  padding: 10px 25px;
```

```
  border-radius: 30px;
  transition: 0.3s;
}

.btn-custom:hover {
  transform: scale(1.08);
}

/* Iframe */
iframe {
  border-radius: 15px;
  box-shadow: 0 8px 20px rgba(0,0,0,0.2);
  transition: 0.3s;
}

iframe:hover {
  transform: scale(1.02);
}

footer {
  background: #0d6efd;
  color: white;
  text-align: center;
  padding: 15px;
}
</style>
</head>

<body>

<!-- NAVBAR -->
<nav class="navbar navbar-expand-lg fixed-top">
 <div class="container">
  <a class="navbar-brand" href="#home">🏠 Housing Trends</a>
  <div>
```

```
      <a class="nav-link d-inline" href="#home">Home</a>
      <a class="nav-link d-inline" href="#dashboard">Dashboard</a>
      <a class="nav-link d-inline" href="#story">Story</a>
    </div>
  </div>
</nav>

<!-- HOME -->
<section id="home" class="home show d-flex align-items-center">
<div class="container">
    <h1 class="display-4 fw-bold" id="title"></h1>
    <p class="lead mt-3">Interactive Tableau Dashboard & Story using Flask</p>

    <a href="#dashboard" class="btn btn-light btn-custom mt-4">
        View Dashboard ↓
    </a>
</div>
</section>

<!-- DASHBOARD -->
<section id="dashboard" class="bg-light">
<div class="container text-center">
    <h2 class="mb-4">📊 Housing Market Dashboard</h2>

    <iframe
src="https://public.tableau.com/views/Book1_17696952321770/Dashboard1?:showVizHome=no&:embed=tru
e"
    width="100%"
    height="700"
    frameborder="0"
    allowfullscreen>
    </iframe>

    <br><br>
```

```
  <a href="#home" class="btn btn-primary btn-custom">Home</a>

  <a href="#story" class="btn btn-success btn-custom">Next → Story</a>

</div>

</section>


<!-- STORY -->

<section id="story" class="bg-white">

<div class="container text-center">

  <h2 class="mb-4">📖 Housing Market Story</h2>


  <iframe

  src="https://public.tableau.com/views/Book1_17696952321770/Story1?:showVizHome=no&:embed=true"

  width="100%"

  height="700"

  frameborder="0"

  allowfullscreen>

  </iframe>


  <br><br>


  <a href="#home" class="btn btn-dark btn-custom">Back to Home</a>

</div>

</section>


<footer>

© 2026 Internship Project | Tableau + Flask

</footer>


<script>


/* Scroll Animation */

const sections = document.querySelectorAll("section");


window.addEventListener("scroll", () => {

  const trigger = window.innerHeight * 0.85;
```

```
  sections.forEach(sec => {
    if(sec.getBoundingClientRect().top < trigger){
      sec.classList.add("show");
    }
  });
});

/* Typewriter Effect */
const text = "Visualizing Housing Market Trends";
let i = 0;
function typing(){
  if(i < text.length){
    document.getElementById("title").innerHTML += text.charAt(i);
    i++;
    setTimeout(typing, 50);
  }
}
typing();

</script>

</body>
</html>
```

- **Dataset Link:**

❖ Kaggle Housing Dataset:

Visualizing Housing Market Trends An Analysis of Sale Prices and Features using Tableau

https://www.kaggle.com/datasets/rituparnaghosh18/transformed-housing-data-2

GitHub & Project Demo Link:

❖ GitHub link:

https://github.com/Lakshmi130604/Visualizing-Housing-Market-Trends-An-Analysis-of-Sale-Prices-and-Features-using-Tableau/tree/main

❖ DEMO VIDEO LINK:

https://drive.google.com/file/d/1df7i8HB70rKN_WyNZiaY7N8xr1_j21z0/view?usp=sharing

❖ DASHBORDLINK:
https://public.tableau.com/views/Book1_17696952321770/Dashboard1?:showVizHome=no&:embed=true

❖ STORY LINK:
https://public.tableau.com/views/Book1_17696952321770/Story1?:showVizHome=no&:embed=true

- Project Web Interface LINK:
  http://127.0.0.1:5000